

Mathematics & Statistics

The Mathematical Brain Behind Every DS Model

■ MATH & STATISTICS

■ LEARNING OUTCOME

By the end of this PDF you will understand the core mathematical and statistical concepts that power every data science algorithm — linear algebra, calculus intuition, probability, and inferential statistics.

■ Skills You Will Gain

- Understand vectors, matrices, and dot products
- Grasp gradient descent intuition from calculus
- Apply probability theory to real-world events
- Use Bayes' Theorem for conditional reasoning
- Calculate and interpret descriptive statistics
- Run and interpret hypothesis tests (t-test, chi-square)

■ Maths is the language of ML models. You don't need to derive proofs — but you MUST understand what each formula does to your data.

SECTION 1

Linear Algebra Essentials

Concept	What It Means	DS Application
Scalar	Single number — e.g. 5.0	A weight or bias in a neuron
Vector	1D array — e.g. [2,3,5]	One data point with multiple features
Matrix	2D grid of numbers	Entire dataset: rows=samples, cols=features
Dot Product	$v1 \cdot v2$ = sum of products of matching elements	Similarity between two vectors
Matrix Multiply	$(m \times n) \times (n \times p) = (m \times p)$	Neural network layer transformation
Transpose	Flip rows and columns: A^T	Swap dimensions for computation
Inverse	$A \times A^{-1}$ = Identity matrix	Solve linear equations

```
import numpy as np
A = np.array([[1,2],[3,4]])
B = np.array([[5,6],[7,8]])
np.dot(A,B) # Matrix multiplication
A.T # Transpose
np.linalg.inv(A) # Inverse
np.linalg.det(A) # Determinant
np.linalg.eig(A) # Eigenvalues & eigenvectors (used in PCA)
```

SECTION 2

Calculus Intuition for DS

Why Calculus?

Machine learning models learn by minimizing a loss function. The derivative tells us the SLOPE — which direction to move weights to reduce error. Gradient Descent uses this slope to update model parameters step by step.

1

Derivative

Rate of change: how much output changes per unit input change — the 'slope'

2

Partial Derivative

Derivative of a function with multiple variables, keeping others constant

3

Gradient

Vector of all partial derivatives — points in direction of steepest increase

4

Gradient Descent

Move weights opposite to gradient to minimize loss: $w = w - \alpha \nabla L$

5

Learning Rate (α)

Step size — too big = overshoot, too small = slow convergence

SECTION 3

Probability Theory

Concept	Formula	Example
Basic Probability	$P(A) = \text{favourable} / \text{total}$	$P(\text{heads}) = 1/2$
AND Rule	$P(A \cap B) = P(A) \times P(B)$ if independent	$P(\text{two heads}) = 0.5 \times 0.5 = 0.25$
OR Rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	$P(\text{head OR tail}) = 1$
Conditional	$P(A B) = P(A \cap B) / P(B)$	$P(\text{sick} \text{test+}) = \text{sensitivity}$
Complement	$P(\text{not } A) = 1 - P(A)$	$P(\text{not rain}) = 1 - 0.3 = 0.7$

SECTION 4

Bayes' Theorem

Bayes' Theorem Formula

$P(A|B) = P(B|A) \times P(A) / P(B)$ Example — Spam Filter: $P(\text{spam} | \text{word}=\text{'free'}) = P(\text{word}=\text{'free'} | \text{spam}) \times P(\text{spam}) / P(\text{word}=\text{'free'})$ Prior $\times$ Likelihood / Evidence = Posterior probability

```
# Naive Bayes in sklearn (uses Bayes' Theorem under the hood) from sklearn.naive_bayes import GaussianNB model = GaussianNB() model.fit(X_train, y_train)
```

SECTION 5

Key Probability Distributions

	Shape	DS Use Case
	Bell curve — symmetric	Most ML assumptions, height, errors
	Discrete — count of successes	Click-through rate, coin flips
	Count of events in time/space	Website visits per hour, defects per item
	All values equally likely	Random number generation, initialization
	Time between events	Customer inter-arrival time

SECTION 6

Inferential Statistics

1	Confidence Interval Range where true parameter lies with X% confidence: $x \pm z^*(\sigma/\sqrt{n})$
2	Null Hypothesis (H₀) Assume no effect — 'new model performs same as baseline'

3	Alternative (H_a) What you want to prove — 'new model is better'
4	p-value Probability of seeing this result by chance if H ₀ is true. $p < 0.05$ = reject H ₀
5	t-test Compare means of two groups — are they significantly different?
6	Chi-square Test independence between two categorical variables
7	ANOVA Compare means across 3+ groups simultaneously

```
from scipy import stats # t-test: are two groups different? t, p = stats.ttest_ind(group_A,
group_B) print(f't={t:.3f}, p={p:.4f}') # Chi-square: are two categories independent? chi2, p,
dof, expected = stats.chi2_contingency(contingency_table) # ANOVA: 3+ groups f, p =
stats.f_oneway(group1, group2, group3)
```

■ PRACTICE QUESTIONS

Q1. What is the shape of a matrix that multiplies (4×3) with (3×5)?

Answer: (4×5) — the inner dimensions must match; result shape is (outer row, outer col).

Q2. In gradient descent, what does the learning rate control?

Answer: The step size when updating weights. Too high = overshoot minimum; too low = very slow training.

Q3. A disease affects 1% of population. A test is 99% accurate. You test positive — what's the chance you're sick?

Answer: Using Bayes' Theorem, only ~50%. The low base rate (1%) means many false positives despite high accuracy.

Q4. What does p-value = 0.02 mean when testing if a new drug works?

Answer: There's only a 2% chance of seeing this result by luck if the drug had no effect. We reject H₀ — the drug likely works.

Q5. What is the difference between variance and standard deviation?

Answer: Variance is the average squared deviation from the mean. Std dev is the square root of variance — same units as the data.

■ Maths & Stats Cheat Sheet	
<code>np.dot(A,B)</code>	Matrix multiplication
<code>A.T</code>	Matrix transpose
<code>np.linalg.inv(A)</code>	Matrix inverse
<code>np.linalg.eig(A)</code>	Eigenvalues (PCA)
$P(A B) = P(B A)P(A)/P(B)$	Bayes' Theorem
$p < 0.05$	Statistically significant
<code>stats.ttest_ind(a,b)</code>	Two-sample t-test
<code>stats.chi2_contingency(t)</code>	Chi-square test
<code>stats.f_oneway(g1, g2, g3)</code>	ANOVA
<code>np.mean/std/var(x)</code>	Descriptive stats

CI = $\bar{x} \pm z^*(\sigma/\sqrt{n})$	Confidence interval
Gradient = $-\alpha \nabla L$	Weight update rule